

The Local Liakopoulos–Meyer Conjecture for Homothetic Fiber Profiles

Partial-result packet for arXiv:2512.02761

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Status: candidate substantial partial result, likely valid; human review requested. The sharp local Liakopoulos–Meyer inequality conjectured in the source holds for every arbitrary s -cover whenever the fiber-volume function has aligned homothetic superlevel sets. This removes the source’s reducibility assumption for a broad class containing nonconstant and nonunconditional bodies, including the first nondecomposable triangle cover. The conjecture for completely arbitrary convex bodies remains open.

1 Source conjecture

For $\sigma \subset [n]$, write $H_\sigma = \text{span}\{e_i : i \in \sigma\}$. A family $(\sigma_1, \dots, \sigma_m)$, with $\sigma_j \subset \sigma$, is an s -cover of σ if every coordinate in σ occurs in exactly s members. Conjecture 1.4 of Alías, González Merino, and Marín Gimeno [1] asserts that every convex body $K \subset \mathbb{R}^n$ satisfies

$$\text{vol}(K)^{m-s} \max_x \text{vol}(K \cap (x + H_\sigma^\perp))^s \geq \frac{\prod_{j=1}^m (|\sigma| - |\sigma_j|)!}{|\sigma|^{m-s}} \prod_{j=1}^m \text{vol}(K \cap H_{\sigma_j}^\perp). \quad (1)$$

Conjecture 1.4. *Let $K \in \mathcal{K}^n$ and $(\sigma_1, \dots, \sigma_m)$ a s -cover of $\sigma \subset [n]$. Then,*

$$\text{vol}(K)^{m-s} \max_{x \in \mathbb{R}^n} \text{vol}(K \cap (x + H_\sigma^\perp))^s \geq \frac{\prod_{j=1}^m (|\sigma| - |\sigma_j|)!}{|\sigma|^{m-s}} \prod_{j=1}^m \text{vol}(K \cap H_{\sigma_j}^\perp).$$

Figure 1: Source Conjecture 1.4 (PDF page 4).

The source proves (1) when the complementary $(m - s)$ -cover is *1-reducible*, meaning that it splits into $m - s$ genuine partitions. The restriction is real: the three edges $\{12, 13, 23\}$ form a 2-cover of $[3]$, but no two edges are disjoint, so this cover cannot split into two 1-covers.

2 A functional arbitrary-cover theorem

Let $p \geq 1$, let $C \subset \mathbb{R}^p$ be a convex body with the origin in its interior, and let $\|\cdot\|_C$ be its gauge. A function $F : \mathbb{R}^p \rightarrow [0, \infty)$ has an *aligned homothetic profile* if

$$F(x) = M\psi(\|x\|_C), \quad (2)$$

where $M > 0$, $\psi : [0, \infty) \rightarrow [0, 1]$ is nonincreasing and right-continuous, $\psi(0) = 1$, and ψ has bounded support. Thus all nonempty superlevel sets of F are dilates of the same C , about the same origin. No symmetry or log-concavity is needed in the next theorem.

Theorem 1 (Arbitrary uniform covers). *Let $(\tau_1, \dots, \tau_m)$ be an r -cover of $[p]$, where $r \geq 1$, and put $d_j = |\tau_j|$. Every F of the form (2) satisfies*

$$M^{m-r} \left(\int_{\mathbb{R}^p} F \right)^r \geq \frac{\prod_{j=1}^m d_j!}{p!^r} \prod_{j=1}^m \int_{H_{\tau_j}} F. \quad (3)$$

There is no reducibility assumption on the cover.

Proof intuition. The common body C and the one-dimensional radial profile ψ can be separated exactly. The geometric factors are governed by Liakopoulos's sharp section inequality for arbitrary covers. The profile factors are moments of one random radius; the cover identity $\sum_j d_j = rp$ makes Lyapunov's moment inequality give exactly the missing comparison, with no loss in the conjectured constant.

Proof. Choose a nonnegative random variable R with survival function $\Pr\{R > t\} = \psi(t)$, and set $\mu_d = \mathbb{E}R^d$, including $\mu_0 = 1$. For every coordinate subspace H_τ of dimension d , Tonelli's theorem gives

$$\begin{aligned} \int_{H_\tau} \psi(\|x\|_C) dx &= \mathbb{E} \text{vol}_d\{x \in H_\tau : \|x\|_C < R\} \\ &= \mu_d \text{vol}_d(C \cap H_\tau). \end{aligned} \quad (4)$$

The formula also holds for $d = 0$ under the usual zero-dimensional volume convention.

Lyapunov's inequality says $\mu_d^{1/d} \leq \mu_p^{1/p}$ for $1 \leq d \leq p$. Since an r -cover satisfies $\sum_j d_j = rp$,

$$\prod_{j=1}^m \mu_{d_j} \leq \mu_p^{\sum_j d_j/p} = \mu_p^r. \quad (5)$$

Zero-dimensional terms equal one and cause no change.

Liakopoulos's dual uniform-cover inequality [2], applied to C and the arbitrary r -cover (τ_j) , is

$$\text{vol}_p(C)^r \geq \frac{\prod_j d_j!}{p!^r} \prod_j \text{vol}_{d_j}(C \cap H_{\tau_j}). \quad (6)$$

By (4), the left side of (3) is $M^m \mu_p^r \text{vol}_p(C)^r$. Combining (5) and (6) bounds it below by

$$\frac{\prod_j d_j!}{p!^r} \prod_j [M \mu_{d_j} \text{vol}_{d_j}(C \cap H_{\tau_j})],$$

which is the right side of (3). □

3 Application to the source conjecture

Let $p = |\sigma|$, $r = m - s$, and $\tau_j = \sigma \setminus \sigma_j$. Then (τ_j) is an r -cover of σ , and $d_j = p - |\sigma_j|$. Define the fiber-volume function

$$F(x) = \text{vol}_{n-p}(K \cap (x + H_\sigma^\perp)), \quad x \in H_\sigma. \quad (7)$$

Corollary 2 (Homothetic fibers). *If the function (7), after identifying H_σ with $\mathbb{R}^p$, has the aligned homothetic form (2), then the full inequality (1) holds for K and every s -cover (σ_j) , whether reducible or not.*

Proof. Fubini's theorem gives

$$\text{vol}_n(K) = \int_{H_\sigma} F, \quad \text{vol}(K \cap H_{\sigma_j}^\perp) = \int_{H_{\tau_j}} F. \quad (8)$$

Moreover $\max F = M$. Substitution of (8) into Theorem 1 is exactly (1). $\square$

The structural hypothesis is realized by a large concrete family. Let $C \subset H_\sigma$ and $B \subset H_\sigma^\perp$ be arbitrary convex bodies containing the origin, put $q = n - p$, and form their gauge ℓ_1 -sum

$$K = C \oplus_1 B := \{(x, y) : \|x\|_C + \|y\|_B \leq 1\}. \quad (9)$$

Its fiber over x is $(1 - \|x\|_C)_+ B$, so

$$F(x) = \text{vol}_q(B)(1 - \|x\|_C)_+^q. \quad (10)$$

Thus Corollary 2 proves (1) for every cover and every body in (9). Neither B , C , nor K need be unconditional, and the fibers are nonconstant when $q > 0$.

For the first nondecomposable case, take $p = 3$, $(\tau_j) = (12, 13, 23)$, $r = 2$, and any $q \geq 1$. The corresponding source cover is $(\sigma_j) = (\{3\}, \{2\}, \{1\})$, with $s = 1$. For (10),

$$\mu_d = \frac{d!q!}{(d+q)!}, \quad \frac{\mu_3^2}{\mu_2^3} = \frac{9}{2} \frac{(q+1)(q+2)}{(q+3)^2} = 1 + \frac{q(7q+15)}{2(q+3)^2} > 1. \quad (11)$$

The exact accompanying script verifies (11) and the cover's non-1-reducibility. This supplies explicit cases not reached by the source's main theorem.

4 Boundary of the result and upgrade audit

The homothetic hypothesis is doing real work. For arbitrary nested convex superlevel sets, applying (6) at each level leaves a reverse-Hölder coupling between different sectional level profiles; standard Hölder and Chebyshev inequalities point in the wrong direction. Reverse Brascamp–Lieb has total cover weight $m/r > 1$, so log-concavity does not compare its supremal convolution with F . A convex-body lift is valid but yields the weaker factorial constant $\prod_j (d_j + q)! (p!)^s / (p + q)!^m$. Finally, the maximizing fiber in (1) need not be centered on the coordinate sections, preventing a direct common translation.

Eight focused attempts covered the exact functional reduction, the triangle cover, polyhedral and ellipsoidal stress tests, level sets, reverse Brascamp–Lieb, convex lifting, the successful moment factorization, and a final Berwald/localization upgrade. No localization preserving all coordinate restrictions supplied the missing coupling. Thus this packet does not claim (1) for arbitrary K .

Verification and novelty. Review should focus on the Fubini identification (8), the random-radius formula (4), and the direction of Lyapunov's inequality in (5). All are exact and the proof contains no numerical step. On 12 August 2026, the four run indexes and exact/current searches for the title, conjecture, arbitrary covers, and homothetic fibers found no later resolution or this structural subcase. Novelty confidence is moderate and provisional.

Human-review recommendation. Review as a substantial partial result. It proves the sharp conjectured constant for all covers in a natural infinite-dimensional parameter family of convex bodies and crosses the source’s combinatorial reducibility barrier, while leaving the unrestricted convex-body problem open.

References

- [1] L. J. Alías, B. González Merino, and B. Marín Gimeno, *On local Liakopoulos–Meyer type inequalities and their functional counterparts*, arXiv:2512.02761, Conjecture 1.4.
- [2] D. M. Liakopoulos, *Reverse Brascamp–Lieb inequality and the dual Bollobás–Thomason inequality*, Arch. Math. 112 (2019), no. 3, 293–304.